

HW #1

①

1. $\vec{r} = (R+x) \hat{i} + y \hat{j} + z \hat{k}$

$$\begin{aligned} (a) \quad \dot{\vec{r}} &= \dot{x} \hat{i} + \dot{y} \hat{j} + \dot{z} \hat{k} + \vec{\omega} \times [(R+x) \hat{i} + y \hat{j} + z \hat{k}] \\ &= \dot{x} \hat{i} + \dot{y} \hat{j} + \dot{z} \hat{k} + \omega \hat{k} \times [(R+x) \hat{i} + y \hat{j} + z \hat{k}] \\ &= [\dot{x} - \omega y] \hat{i} + [\dot{y} + \omega(R+x)] \hat{j} + \dot{z} \hat{k} \end{aligned}$$

$$\begin{aligned} \ddot{\vec{r}} &= [\ddot{x} - \omega \dot{y}] \hat{i} + [\ddot{y} + \omega \dot{x}] \hat{j} + \ddot{z} \hat{k} \\ &\quad + \vec{\omega} \times [\dot{x} \hat{i} + \dot{y} \hat{j} + \dot{z} \hat{k}] \\ &= [\ddot{x} - 2\omega \dot{y} - \omega^2(R+x)] \hat{i} + [\ddot{y} + 2\omega \dot{x} - \omega^2 y] \hat{j} + \ddot{z} \hat{k} \end{aligned}$$

$$(b) \quad \begin{cases} \ddot{x} - 2\omega \dot{y} - \omega^2(R+x) = -\frac{\mu}{r^3}(R+x) + u_x \\ \ddot{y} + 2\omega \dot{x} - \omega^2 y = -\frac{\mu}{r^3} y + u_y \\ \ddot{z} = -\frac{\mu}{r^3} z + u_z \end{cases}$$

$$(c) \quad \bar{x} = \frac{x}{R}, \quad \bar{y} = \frac{y}{R}, \quad \bar{z} = \frac{z}{R}, \quad \bar{t} = \omega t$$

$$\begin{cases} \ddot{\bar{x}} - 2\dot{\bar{y}} - (1+\bar{x}) = -\frac{1+\bar{x}}{\bar{r}^3} + u_x \\ \ddot{\bar{y}} + 2\dot{\bar{x}} - \bar{y} = -\frac{\bar{y}}{\bar{r}^3} + u_y \\ \ddot{\bar{z}} = -\frac{\bar{z}}{\bar{r}^3} + u_z \end{cases}$$

$$\bar{r} = \sqrt{(1+\bar{x})^2 + \bar{y}^2 + \bar{z}^2}$$

Remove '-' for simplicity to yield

$$\left. \begin{aligned} \ddot{x} - 2\dot{y} - (1+x) &= -\frac{1+x}{r^3} + u_x \\ \ddot{y} + 2\dot{x} - y &= -\frac{y}{r^3} + u_y \\ \ddot{z} &= -\frac{z}{r^3} + u_z \end{aligned} \right\}$$

$$r = \sqrt{(1+x)^2 + y^2 + z^2}$$

(d) Operating Point $x=y=z \equiv 0$.

~~$$(x = 0 + \Delta x, y = 0 + \Delta y, z = 0 + \Delta z)$$~~

Introduce $\begin{cases} x \triangleq 0 + \Delta x \\ y \triangleq 0 + \Delta y \\ z \triangleq 0 + \Delta z \end{cases}$

and lineare ...

$$\begin{cases} \Delta \ddot{x} - 2\Delta \dot{y} - 3\Delta x = u_x \\ \Delta \ddot{y} + 2\Delta \dot{x} = u_y \\ \Delta \ddot{z} + \Delta z = u_z \end{cases}$$

Remove 'Δ' for simplicity to yield.

$$\left(\begin{aligned} \ddot{x} - 2\dot{y} - 3x &= u_x \\ \ddot{y} + 2\dot{x} &= u_y \\ \ddot{z} + z &= u_z \end{aligned} \right)$$

(3)

$$2. \quad \ddot{r} - r\dot{\nu}^2 \cos^2 \lambda - r\dot{\lambda}^2 = -\frac{\mu}{r^2} + \frac{F_r}{m} \quad (1)$$

$$\ddot{\nu} \cos \lambda + 2\dot{r}\dot{\nu} \cos \lambda - 2r\dot{\nu}\dot{\lambda} \sin \lambda = \frac{F_\nu}{m} \quad (2)$$

$$r\ddot{\lambda} + 2\dot{r}\dot{\lambda} + r\dot{\nu}^2 \sin \lambda \cos \lambda = \frac{F_\lambda}{m} \quad (3)$$

$$\mu_E = 3.99 \times 10^{14} \text{ m}^3/\text{s}^2$$

$$3. (a) \quad r = R_E + 600 \text{ km} = 6.37 \times 10^3 + 6 \times 10^2$$

$$= \underline{\underline{6.97 \times 10^3}}$$

$$r = R_E + 600 \text{ km} = 6.37 \times 10^3 + 6 \times 10^2$$

$$= 6.97 \times 10^3 \text{ km}$$

$$\dot{\nu} = \text{constant} = \omega_0$$

$$\lambda = 0 \Rightarrow \dot{\lambda} = 0$$

Introducing to (1) above,

$$0 - r\omega_0^2 - r \cdot 0^2 = -\frac{\mu}{r^2} + 0$$

$$\Rightarrow \omega_0 = \sqrt{\frac{\mu}{r^3}} = 1.09 \times 10^{-3} \text{ rad/sec}$$

Constant Orbital Velocity

$$V = r\omega_0 = \sqrt{\frac{\mu}{r}} = 7.56 \text{ km/sec}$$

$$(b) \quad T = \frac{2\pi}{\omega_0} = 5.8176 \times 10^3 \text{ sec}$$

$$= \underline{\underline{1.62 \text{ hr}}}$$

$$= \underline{\underline{1.616 \text{ hr}}}$$

4.

$$\begin{cases} \ddot{x} - 2\omega_0 \dot{y} - 3\omega_0^2 x = \frac{F_r}{m} \\ \ddot{y} + 2\omega_0 \dot{x} = \frac{F_v}{m} \\ \ddot{z} + \omega_0^2 z = \frac{F_a}{m} \end{cases} \quad \begin{cases} x \triangleq \Delta R \\ y \triangleq R \Delta v \\ z \triangleq R \Delta \lambda \end{cases}$$

$$r = R_E + 60 \text{ km} = 6.97 \times 10^3 \text{ km}$$

$$\omega_0 = \sqrt{\frac{\mu}{r^3}} = 1.09 \times 10^{-3} \text{ rad/sec.}$$

5 (a) The solutions to n-plane CW eqn TS.

$$\begin{bmatrix} x(t) \\ v_x(t) \\ y(t) \\ v_y(t) \end{bmatrix} = \begin{bmatrix} \text{STM} \end{bmatrix} \begin{bmatrix} x(0) \\ v_x(0) \\ y(0) \\ v_y(0) \end{bmatrix}$$

Operating Point $(r, v, \lambda) = (R, w_0, 0)$

$$R = 6.37 \times 10^3 + 6 \times 10^2 = 6.97 \times 10^3 \text{ km}$$

$$\omega_0 = \sqrt{\frac{\mu}{R^3}} = 1.09 \times 10^{-3} \text{ rad/sec}$$

Tangential Velocity (before) perturbation is

$$V = R\omega_0 = 7.56 \text{ km/sec}$$

Then the initial conditions are

$$\begin{aligned} x(0) &= v_x(0) = y(0) = 0 \\ v_y(0) &= 0.01 R\omega_0 = 7.56 \times 10^{-2} \text{ km/sec} \end{aligned}$$

(5)

Introducing the IC to STM.

$$\begin{bmatrix} x(t) \\ v_x(t) \\ y(t) \\ v_y(t) \end{bmatrix} = \begin{bmatrix} \frac{2C(1-\cos\omega_0 t)}{\omega_0} \\ 2Sm\omega_0 t \\ \frac{4Sm\omega_0 t}{\omega_0} - 3t \\ -3 + 4\cos\omega_0 t \end{bmatrix} V_y(0)$$

$$\begin{cases} r(t) = R + \frac{2(1-\cos\omega_0 t)}{\omega_0} V_y(0) \\ v(t) = \omega_0 t + \frac{1}{R} \left(\frac{4Sm\omega_0 t}{\omega_0} - 3t \right) V_y(0) \\ \lambda(t) = 0 \end{cases} \quad (1)$$

$$\begin{cases} \dot{r}(t) = 2Sm\omega_0 t \cdot V_y(0) \\ \dot{v}(t) = \omega_0 + \frac{1}{R} (4\cos\omega_0 t - 3) V_y(0) \\ \dot{\lambda}(t) = 0 \end{cases}$$

(b) Orbital Position Vector.

$$\vec{r}(t) = (7.11 \times 10^3 - 1.39 \times 10^2 \cos\omega_0 t) \vec{e}_r \text{ km}$$

$$r_p = 7.11 \times 10^3 - 1.39 \times 10^2$$

$$= 6.97 \times 10^3 \text{ km/sec}$$

$$r_a = 7.11 \times 10^3 + 1.39 \times 10^2$$

$$= 7.25 \times 10^3 \text{ km/sec}$$

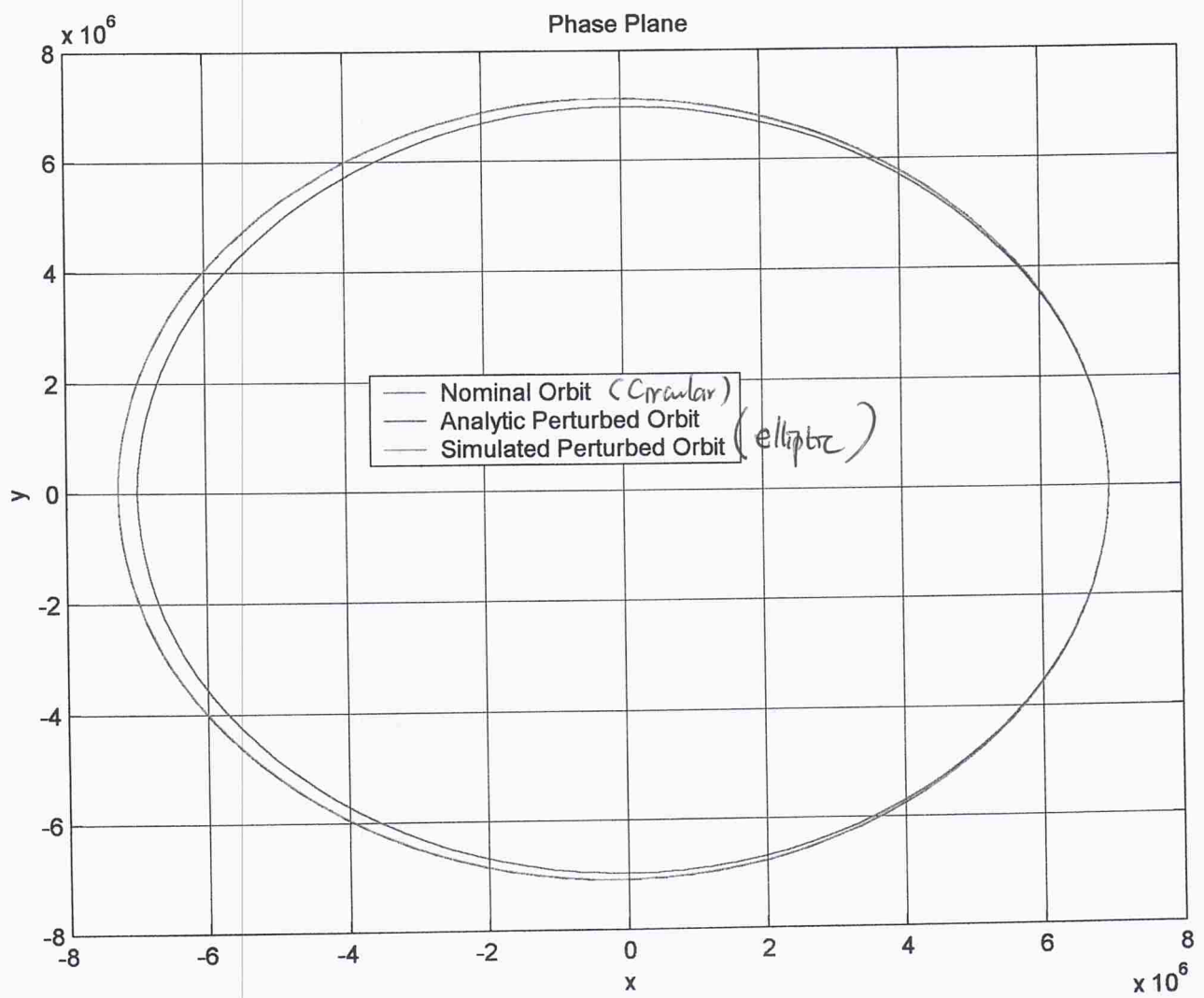
$$e = \frac{r_a - r_p}{r_a + r_p} = 0.0197$$

$$a = \frac{r_a + r_p}{2} = 7.118 \times 10^3 \text{ km}$$

$$T = 2\pi \sqrt{\frac{a^3}{\mu}} = 5.9761 \times 10^3 \text{ sec}$$

$$= 1.66 \text{ hr}$$

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$$6. (a) A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 3\omega_0^2 & 0 & 0 & 2\omega_0 \\ 0 & 0 & 0 & 1 \\ 0 & -2\omega_0 & 0 & 0 \end{bmatrix} \quad b_1 = \begin{bmatrix} 0 \\ \frac{1}{m} \\ 0 \\ 0 \end{bmatrix}$$

$$\text{rank}([b_1 \quad Ab_1 \quad A^2b_1 \quad A^3b_1]) = 3$$

$\Rightarrow$ Not completely controllable

$$(b) A = \begin{bmatrix} \dots \end{bmatrix} \quad b_2 = \begin{bmatrix} 0 \\ 0 \\ 0 \\ \frac{1}{m} \end{bmatrix}$$

$$\text{rank}([b_2 \quad Ab_2 \quad A^2b_2 \quad A^3b_2]) = 4$$

$\Rightarrow$ Completely Controllable

$$(c) A = \begin{bmatrix} \dots \end{bmatrix} \quad B = \begin{bmatrix} b_1 & b_2 \\ \vdots & \vdots \end{bmatrix}$$

$$\text{rank}([B \quad AB \quad A^2B \quad A^3B]) = 4$$

$\Rightarrow$ Completely Controllable

⇒ Start here!

(8)

$$7. \quad r_1 = (6.378 \times 10^3 + 5.80 \times 10^2) \text{ km}$$

$$r_2 = (6.378 \times 10^3 + 6.00 \times 10^2) \text{ km}$$

$$t_f = 30 \text{ min.}$$

(a) Ref. Orbit $r = r_2 = 6.978 \times 10^3 \text{ km}$

$$\omega_0 = \sqrt{\frac{\mu}{r_2^3}} = 1.09 \times 10^{-3} \text{ rad/sec}$$

$$\begin{matrix} t = t_0 \\ \begin{bmatrix} x_0 \\ \dot{x}_0 \\ y_0 \\ \dot{y}_0 \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} t = t_0^+ \\ \begin{bmatrix} R_1 - R_2 = -20 \text{ km} \\ 0 \\ 0 \\ -\frac{3}{2} \omega_0 x_0 = 0.0327 \text{ km/sec} \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} t = t_0^+ \\ \begin{bmatrix} -20 \text{ km} \\ \Delta V_x^+ \\ 0 \\ 0.0327 \text{ km/sec} + \Delta V_y^+ \end{bmatrix} \end{matrix}$$

$$\begin{matrix} \text{CW} \\ \text{STM} \end{matrix} \rightarrow \begin{matrix} t = t_f^- \\ \begin{bmatrix} f(\Delta V_x^+, \Delta V_y^+) \end{bmatrix} \end{matrix} \rightarrow \begin{matrix} t = t_f \\ \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{matrix}$$

~~$$\Delta \vec{V}_0 = \Delta V_x^+ \hat{x} + \Delta V_y^+ \hat{y}$$~~

$$\begin{aligned} \Delta \vec{V}_0 &= \Delta V_x^+ \hat{x} + \Delta V_y^+ \hat{y} \\ &= 0.023053 \text{ km/sec } \hat{x} + 0.0000 \text{ km/sec } \hat{y} \end{aligned}$$

Need to be checked

$$\begin{aligned} \Delta \vec{V}_f &= \Delta V_x^- \hat{x} + \Delta V_y^- \hat{y} \\ &= 0.0084 \text{ km/sec } \hat{x} + 0.0107 \text{ km/sec } \hat{y} \end{aligned}$$

Need to be checked!

(b)

$$\begin{bmatrix} x \\ v_x \\ y \\ v_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

~~30 min~~
~~30 min~~
 $t_f = 30 \text{ min}$

$$= e^{At_f} \begin{bmatrix} -20 \text{ km} \\ 0 \\ 0 \\ 0.0327 \text{ km/sec} \end{bmatrix} + \int_0^{t_f} e^{A(t_f-\tau)} B d\tau \begin{bmatrix} T_r^1 \\ T_v^1 \end{bmatrix} + \int_{t_1}^{t_f} e^{A(t_f-\tau)} B d\tau \begin{bmatrix} T_r^2 \\ T_v^2 \end{bmatrix}$$

Let $t_f = 15 \text{ min}$.

Numerical Computation gives the value of

$$\begin{bmatrix} T_r^1 \\ T_v^1 \\ T_r^2 \\ T_v^2 \end{bmatrix} = \dots$$

Need to be evaluated

Verify by Simulation.